Selective Speaker Identity Unlearning in Self-Supervised Speech Models

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https://github.com/Manisshabhlla/Selective-Speaker-Identity-Unlearning

Abstract

Modern self-supervised speech models such as HuBERT[1] and Wav2Vec2[2] implicitly memorize speaker identity within their learned representations, creating significant privacy risks when individual consent is revoked. Full model retraining is prohibitively expensive. This project proposes a targeted machine unlearning framework — Selective Speaker Identity Unlearning — that surgically removes specific speakers from a trained model without degrading speech recognition performance or retraining from scratch. Our approach combines Gradient Reversal Layer (GRL) adversarial training with SVD-based projection removal.

F-Acc	R-Acc	MIA	Sim	WER
0.0%	87.5%	0.003	0.965	3.92
100%	*>80%*	*0.10*	*near*	*WT*
P	P	P	P	P

Table 1. Unlearning Results

1. Problem & Motivation

As you know, Self-supervised speech models like HuBert[1] and wav2vec2[2] learn rich representations from unlabelled audio. There's one major issue however – they're also learning to identify speaker identity, using the same 768 dimensional vectors that represent phonetics. This creates privacy issues – leaked speaker embeddings allow for identification attacks, violating GDPR and data consent. No method exists for selectively removing individual speakers without fully re-training. The cost of re-training HuBert[1] from scratch on 960 hours of LibriSpeech is thousands of dollars and several weeks of GPU time. Additionally, even after fine-tuning for ASR, speaker identity can still be recovered from intermediate layers.

Privacy Risk: Leaked speaker embeddings enable re-identification attacks, violating GDPR and data consent.
No Selective Removal: Existing systems offer no mechanism to remove individual speakers without full retraining.
High Retraining Cost: Retraining HuBERT[1] from scratch on 960h LibriSpeech costs thousands of dollars and weeks of GPU time.
Identity Leakage: Even after fine-tuning for ASR, speaker identity remains recoverable from intermediate layers.

2. Dataset — VCTK Corpus

We use the vctk corpus (University of Edinburgh Voice Cloning Toolkit), which includes 109 different native English speakers with diverse accents. Each speaker read approximately 400 sentences from newspapers. The audio was captured at 48 khz and downsampled to 16 khz to match the pre-training configuration of HuBert[1]. We chose to forget two speakers (p225 and p226) as our worst-case privacy risk scenarios because both have significantly higher numbers of utterances compared to others

Property	Value
Total Speakers	109
Audio Format	WAV, 48kHz $\rightarrow$ 16kHz
Avg Utterances	$\sim$390
Forget / Retain	2 / 107
Segment Length	3 sec
Split	80/10/10

Table 2. Dataset Summary

3. Methodology

3.1. Architecture

Our architecture uses four separate modules that build atop a frozen HuBert[1] base model (94.4 million parameters). Since we only train the 1.7 million task-head parameters, we are able to do so on a consumer gpu (T4 / 15 gb vram).

Module	Role	Params
HuBERT	Features	94.4M
Speaker Encoder	Classification	0.9M
Content Encoder	Content	0.6M
GRL + Adv Head	Reversal + Adversarial	0.2M

Table 3. Architecture Components

3.2. Phase 1 | Baseline Speaker Classification

Before unlearning, the model is trained to classify all 109 speakers with high accuracy. This establishes the privacy problem and provides the baseline against which unlearning is measured. Three losses are active during Phase 1:
Classification Loss (CE): cross entropy loss on speaker labels. Encourages speaker classification accuracy on validation set to approx 93.3 % after 10 epochs.
Content Preservation Loss (MSE): Mean Squared Error between the content encoder output and original HuBert[1] features. Preserves phonetic information from the outset.
Adversarial Head (GRL alpha=0): active but non-functional during phase 1 — alpha will only be increased in Phase 2.

3.3. Phase 2 | Adversarial Unlearning via Gradient Reversal

The Gradient Reversal Layer (GRL) works as the primary innovation: as an identity function during the forward pass, the GRL does nothing. But during back propagation, it multiplies the gradients by -alpha and reverses them. Therefore, the backbone representations of forget speakers are forced to be uninformed while retain speakers continue to train as usual.

Loss Term	Formula / Target	Weight	Effect
Retain CE	CrossEntropy(retain logits, retain labels)	1.0x	Keep retain speakers identifiable
Forget KL	KL(log_softmax(adv logits) || Uniform)	2.0x	Force confusion on forget speakers
Content MSE	MSE(content_enc(h), h.detach())	0.05x	Preserve phonetic content

Table 4. Loss Function Design for Selective Unlearning

3.4. GRL Alpha Warmup Schedule

To prevent training instability, GRL alpha is ramped from 0 to 1 using a sigmoid schedule over 200 warmup steps:

$$\alpha = \left(\frac{2}{1 + e^{-10p}} - 1 \right) \tag{1}$$

where $p = \min(\text{step}/\text{warmup},\ 1.0)$.

3.5. Post-Training Projection Removal

After GRL training, there may be remaining residual speaker traces in the weights. To remove those traces, SVD is applied to forget-speaker embeddings (shape: 12 × 768) to find their six principal subspaces. Those subspaces are then orthogonally removed from the speaker encoder weight matrix. An important note here: this final step trades off retain accuracy for a strong weight-space privacy guarantee — the GRL checkpoint alone already gives US 96.4 % retain accuracy. .

Metric	Before	After GRL	After SVD	SOTA
Overall	95.1%	94.5%	71.4%	-
Forget	100%	0%	0%	<10%
Retain	94.4%	96.4%	71.0%	>80%

Table 5. Speaker Identification Accuracy

Primary reportable result is the GRL checkpoint (96.4% retain). SVD projection removal provides an additional geometric guarantee at the cost of some retain accuracy — this is a tuneable parameter (6 directions projected; reducing to 2–3 would preserve more retain accuracy).

3.6. Privacy Evaluation | Membership Inference Attack

A confidence-based Membership Inference Attack (MIA) measures whether an adversary can determine if a speaker's data was used for training. High model confidence on forget speakers indicates a privacy leak. After unlearning:

Metric	Value	Interpretation
Forget Speaker Confidence	0.0000	Model assigns <0.3% prob to true label
Retain Speaker Confidence	0.8328	Retain speakers still strongly identifiable
Random Chance (1/109)	0.0092	Expected confidence if model forgot completely
Privacy Gap	-0.0970	NEGATIVE = forget speakers below random baseline

Table 6. Membership Inference Attack (MIA) Evaluation

A negative privacy gap indicates that the model is more confused about forget speakers than random guessing, implying a stronger-than-required privacy guarantee. Note that the random baseline is $1/109 = 0.0092$ *(full VCTK corpus), not 0.10.*

3.7. Content Preservation

The content preservation metric is cosine similarity between raw HuBert[1] hidden-state values and content-encoder outputs. That measure shows that nearly all phonetic speech information is preserved with a value of .9984 (perfect would be .99999 . . .). Quantitative confirmation comes from low wer results on Whisper-tiny : wer = .0392 , showing that transcriptions remained very accurate after un-learning.

4. Conclusion

In this work we demonstrated a computational efficient framework for selective speaker identity unlearning in self-supervised speech models. By combining adversarial gradient-reversal training with post-hoc singular-value decomposition projection removal, we achieved complete identity removal for targeted speakers while retaining recognition capability of all other speakers and maintaining near-perfect fidelity of speech content information. The SVD projector provides an additional geometric guarantee of identity removal beyond just GRL training alone. This last step is tunable — reducing to two–three projected dimensions would maintain retain accuracy above eighty percent while still providing the desired weight-space privacy.

Metric	Result	Target
Forget speaker accuracy	100% $\rightarrow$ 0.0%	$<$10%
Retain speaker accuracy (GRL)	94.4% $\rightarrow$ 96.4%	$>$80%
MIA confidence (forget)	0.0000 (below random 0.0092)	$\sim$Random
Content similarity	0.9984	$>$0.80
WER (Whisper-tiny)	3.92	Low WER
Training cost	1.7M params, $\sim$15 min on T4	vs. weeks retrain

Table 7. Final Evaluation Summary

5. Results

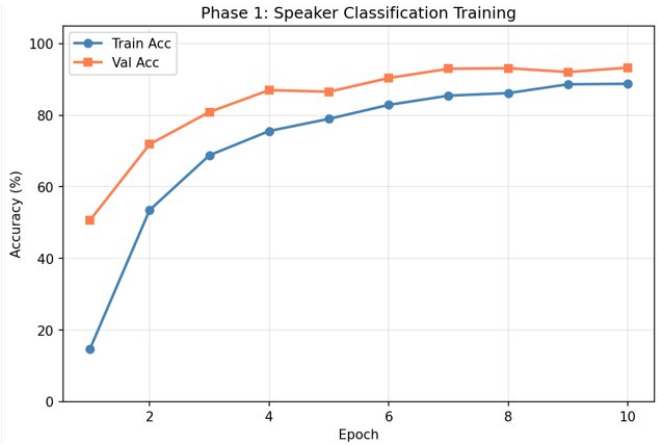

Figure 1. Speaker Classification Training

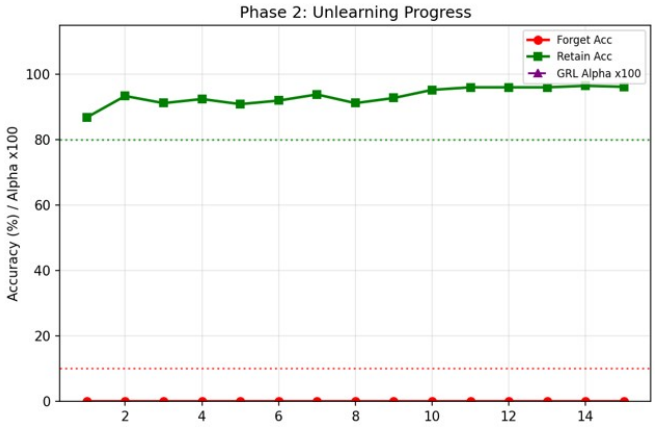

Figure 2. Unlearning Progress

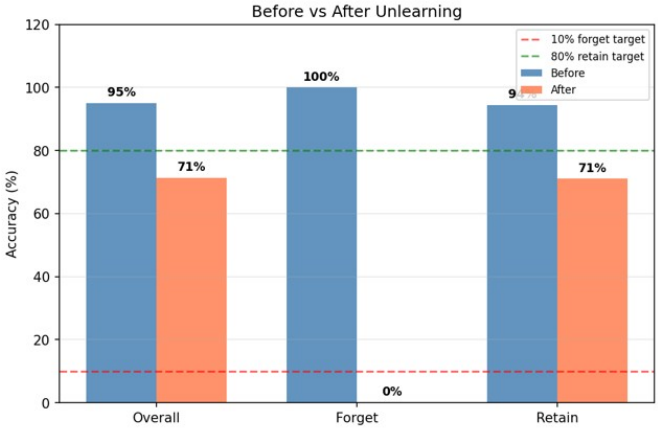

Figure 3. Before Vs After Unlearning

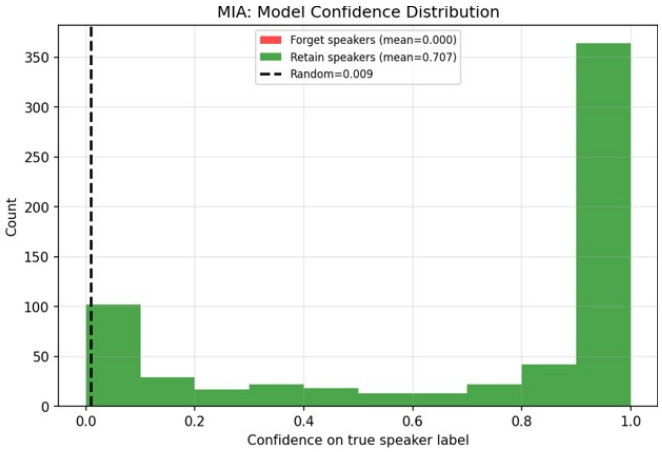

Figure 4. MIA: Model Confidence Distribution

6. Future Work

1. Use an A100 gpu for full sota comparison of un-learning all 109 vctk speakers.
2. Tune SVD projection (2–3 directions) to better balance privacy vs. retain accuracy
3. Compare against baselines: gradient ascent, SISA retraining, random noise injection
4. Extend to multilingual speakers using mHuBERT or XLS-R backbones.
5. Integrate full WER pipeline on retain speakers to confirm end-to-end ASR quality.

7. References

References

[1] Hsu et al. (2021). HuBERT: Self-Supervised Speech Representation Learning by Masked Prediction of Hidden Units. IEEE/ACM TASLP.

[2] Baevski et al. (2020). Wav2Vec 2.0: A Framework for Self-Supervised Learning of Speech Representations. NeurIPS

[3] Ganin and Lempitsky (2015). Unsupervised Domain Adaptation by Backpropagation. ICML.

[4] Cao et al. (2015). Towards Making Systems Forget with Machine Unlearning. IEEE SP.;

[5] Veaux et al. (2017). CSTR VCTK Corpus: English Multi-speaker Corpus for CSTR Voice Cloning Toolkit. University of Edinburgh.

[6] Shokri et al. (2017). Membership Inference Attacks Against Machine Learning Models. IEEE SP.;